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of dead RBCs, bile salts, cholesterol and phospholipids but has no enzymes. Bile helps in emulsification of fats. Bile salts reduce the surface tension of fat droplets and break them into small globules. Bile also activates lipases to digest lipids.

Proteins and partially digested proteins in the chyme on reaching the intestine are acted upon by the proteolytic enzymes of pancreatic juice. Trypsin hydrolyses proteins into polypeptides and peptones, while chymotrypsin hydrolyses peptide bonds associated with specific amino acids.

The pancreatic amylase converts glycogen and starch into maltose. Lipase acts on emulsified fat (triglycerides) and hydrolyses them into free fatty acid and monoglycerides. Monoglycerides are further hydrolysed to fatty acid and glycerol. Nucleases in the pancreatic juice break the nucleic acid into nucleotides and nucleosides.

The secretions of the Brunner's gland along with the secretions of the intestinal glands constitute the intestinal juice or **succus entericus**. The enzymes in the intestinal juice such as maltase, lactase, sucrase (invertase), dipeptidases, lipases, nucleosidases act on the breakdown products of bile and pancreatic digestion.

Maltose $\xrightarrow{\text{Maltase}}$ glucose + glucose

Sucrose $\xrightarrow{\text{sucrase}}$ glucose + fructose

Lactose $\xrightarrow{\text{Lactase}}$ glucose + galactose

Dipeptides, Tripeptides $\xrightarrow{\text{Peptidase}}$ amino acids

Nucleotides $\xrightarrow{\text{Nucleotidase}}$ Nucleoside + Phosphoric acid

Nucleoside $\xrightarrow{\text{Nucleosidase}}$ Sugar + Nitrogen base

Diglycerides and monoglycerides $\xrightarrow{\text{Lipases}}$ Fatty acids + glycerol

The mucus along with the bicarbonate ions from the pancreas provides an alkaline medium (pH 7.8) for the enzymatic action. As a result of digestion, all macromolecules of food are converted into their corresponding monomeric units.

Carbohydrates $\longrightarrow$ monosaccharides (glucose, fructose, galactose)

Proteins $\longrightarrow$ amino acids

Lipids $\longrightarrow$ fatty acids and glycerol

The simple substances thus formed are absorbed in the jejunum and ileum region of the small intestine. The undigested and unabsorbed substances are propelled into the large intestine. The activities of the gastro-intestinal tract are carried out by the neural and hormonal control for proper

Do you feel ill after drinking milk or after eating dairy products?

If so, you cannot digest disaccharide lactose in milk because, the intestinal enzyme lactase is either inactive or absent or present only in very small amounts. The undigested lactose remains in the gut in such persons with lactose intolerance and is broken down by bacteria, causing gas, bloating, stomach cramps and diarrhoea.